

Paper Practice 3.2**Functions**

WORKED KEY – Show all of your work in the space provided. For every yes-or-no question, explain your reasoning in a complete sentence.

- 1.** Determine whether each relation is a function.

Explain. (Relations)

a. $\{(2, 5), (4, 7), (8, 11), (4, 13)\}$

ANSWER: No – the input 4 is paired with both 7 and 13.

b. $\{(-3, 6), (0, 6), (5, 6), (9, 6)\}$ ANSWER:
Yes – every input appears once. A repeated output is allowed.

c. $\{(1, 2), (2, 4), (3, 6), (2, 8)\}$ ANSWER:
No – the input 2 is paired with both 4 and 8.

- 2.** State the domain and the range of each relation. (Domain & Range)

a. $\{(-4, 3), (-1, 0), (2, 5), (6, 3)\}$

ANSWER: $D = \{-4, -1, 2, 6\}$ $R = \{0, 3, 5\}$

b. $\{(0, -2), (1, -2), (2, -2)\}$ ANSWER: $D = \{0, 1, 2\}$ $R = \{-2\}$

- 3.** A mapping pairs $3 \rightarrow 7$, $5 \rightarrow 7$, $8 \rightarrow 9$, and $3 \rightarrow 11$. Is the relation a function? Explain. (Mapping)
ANSWER: No – 3 is paired with both 7 and 11, so one input has two outputs.

- 4.** If $f(x) = 3x - 7$, find each value. (Function Notation)

a. $f(4)$ ANSWER: $f(4) = 3(4) - 7 = 5$

b. $f(-2)$ ANSWER: $f(-2) = 3(-2) - 7 = -13$

c. $f(0)$ ANSWER: $f(0) = -7$

- 5.** If $g(x) = x^2 + 2x$, find each value. (Function Notation)

a. $g(3)$ ANSWER: $g(3) = 9 + 6 = 15$

b. $g(-5)$ ANSWER: $g(-5) = 25 - 10 = 15$

c. $g(0)$ ANSWER: $g(0) = 0$

- 6.** If $f(x) = 2x + 9$, find the value of x for which $f(x) = 21$. (Solve for the Input) ANSWER: $2x + 9 = 21$, so $2x = 12$ and $x = 6$.

7. A vertical line drawn at $x = 2$ crosses a graph at $(2, 1)$ and at $(2, -3)$. Is the graph a function? Explain. (Vertical Line Test) ANSWER: No – it fails the vertical line test. The input 2 has two outputs.

8. Madison took five quizzes. Her scores were Quiz 1: 86, Quiz 2: 92, Quiz 3: 78, Quiz 4: 86, Quiz 5: 84. (Analyze Data)

a. What is the domain? ANSWER: $D = \{1, 2, 3, 4, 5\}$ (the quiz numbers)

b. What is the range? ANSWER: $R = \{78, 84, 86, 92\}$ – 86 is listed once

c. Is this relation a function? Explain. ANSWER: Yes – each quiz number has exactly one score.

9. The table below shows a relation. Is it a function? Explain. (Table)

a. $x: -2, -1, 0, 1$ $y: 5, 5, 5, 5$
ANSWER: Yes – every input appears once. Outputs may repeat.

10. Error Analysis – Gus says $\{(1, 4), (2, 4), (3, 4)\}$ is not a function because the value 4 repeats. Identify his mistake, then state whether the relation is a function. (Error Analysis)
ANSWER: A repeat in the RANGE is allowed; only a repeated input with different outputs breaks a function. It IS a function.

11. Write the relation $\{(1, 3), (2, 6), (3, 9), (4, 12)\}$ as a rule in function notation. (Write a Rule)
ANSWER: $f(x) = 3x$

12. Write a set of four ordered pairs that is NOT a function, then explain in one sentence what makes it fail. (Construct) ANSWER: Answers vary. Any set repeating an input with two different outputs, e.g. $\{(1, 2), (1, 5), (3, 4), (4, 6)\}$ – the input 1 has two outputs.